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Unit – 4 : Session – 7 : SLO – 2 | Binary Search Trees — Deletion Operation

■ I. Fill in the Blanks

- Deletion in BST has THREE cases.
- Case 1: Node to be deleted is a LEAF.
- Case 2: Node has ONE child.
- Case 3: Node has TWO children.
- In case 3, find the INORDER SUCCESSOR (or inorder predecessor).
- In case 3, replace the node with its INORDER SUCCESSOR'S value.
- In case 3, finally DELETE the inorder successor node.
- In case 2, if left child is missing, replace with RIGHT child.
- In case 2, if right child is missing, replace with LEFT child.
- A leaf node can be directly REMOVED (deleted).

■ II. Algorithm to Delete a Node from a BST

- Delete (TREE, VAL)
- Step 1: IF TREE = NULL
- Write "VALUE NOT FOUND"
- ELSE IF VAL < TREE -> DATA
- Delete(TREE -> LEFT, VAL)
- ELSE IF VAL > TREE -> DATA
- Delete(TREE -> RIGHT, VAL)
- ELSE IF TREE -> LEFT AND TREE -> RIGHT
- SET TEMP = findLargestNode(TREE -> LEFT)
- SET TREE -> DATA = TEMP -> DATA
- Delete(TREE -> LEFT, TEMP -> DATA)
- ELSE
- SET TEMP = TREE
- IF TREE -> LEFT = NULL AND TREE -> RIGHT = NULL
- SET TREE = NULL
- ELSE IF TREE -> LEFT != NULL
- SET TREE = TREE -> LEFT
- ELSE
- SET TREE = TREE -> RIGHT
- [END OF IF]
- FREE TEMP
- [END OF IF]
- Step 2: END